

HARSHIT VERMA

Backend Engineer | Python & FastAPI | System Design | Microservices & APIs | AI Productization | React

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SUMMARY

Backend Engineer with hands-on experience designing scalable, high-performance backend systems, REST APIs, and microservices in Python (FastAPI) and Java (Spring Boot), and productizing AI solutions by integrating ML models and LLMs (RAG pipelines, tool-calling agents) into real applications. Built fintech-grade systems including a distributed payment wallet with Saga-based transaction consistency and a high-concurrency booking engine using Redis distributed locks. Completed one year of freelance engineering for MediGlobal, a UK healthcare staffing firm, architecting its core staffing CRM end to end (database, backend, REST APIs, frontend, Azure integration). Experienced in system design, design patterns, performance tuning (bottlenecks and database inefficiencies), and full-stack delivery with React; strong foundation in PostgreSQL, MySQL, Redis, Kafka, and Docker. Currently at Cognizant; pursuing an MCA in AI & ML.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, JavaScript, PHP, PowerShell
- **Backend & APIs:** FastAPI, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, RESTful APIs, Microservices
- **Engineering Practices:** System Design, Scalable Backend Architecture, Design Patterns, Layered Architecture, Performance Optimization, Debugging
- **AI / LLM:** LLM Integration, RAG Pipelines, Embeddings & Vector Search, OpenAI API, Tool Calling / AI Agents, Prompt Engineering, LLM Evaluation
- **Distributed Systems:** Redis (Redlock), Apache Kafka, Saga Pattern, Optimistic & Pessimistic Locking, ACID Transactions, Concurrency Control
- **Databases:** PostgreSQL, pgvector, MySQL, Schema Design, Indexing, Query Optimization, ETL Pipelines
- **Machine Learning:** Scikit-learn, TensorFlow, Keras, NLP, Anomaly Detection, Supervised & Unsupervised ML
- **Security:** JWT Authentication, Spring Security, Role-Based Access Control (RBAC)
- **Cloud & Tools:** Docker, Microsoft Azure, AWS Cloud Fundamentals, Git, GitHub, ServiceNow, Power BI
- **Full-Stack / Frontend:** React.js, JavaScript, HTML, CSS, Tailwind CSS

PROFESSIONAL EXPERIENCE

Analyst – IT Operations & Automation

Cognizant Technology Solutions

📅 Jun 2026 – Present

- Building ServiceNow automation workflows with scripting to streamline access provisioning and cut repetitive Service Desk tasks in the Digital Workspace Services (DWS) team, saving 5+ hours/week.
- Automating routine IT operations with PowerShell and exploring Microsoft Copilot for AI-assisted enterprise support workflows.
- Administering Active Directory and resolving infrastructure incidents (Windows, network, VPN) through structured ticketing within SLA targets.

Freelance Software Engineer – Full-Stack, Backend & Data Science

MediGlobal – Healthcare Staffing & Recruitment, UK (Remote)

📅 2025 – 2026 (1 Year)

Live: crm.mediglobal.info (Staffing CRM) · mediglobal.info (Company Website)

- Sole engineer for a UK healthcare staffing firm placing professionals with NHS and private providers, owning backend, full-stack, and data work across the company's platforms.
- Architected and built the core Medical Staffing CRM end to end — database, backend, REST APIs, and frontend — using PHP, MySQL, React, JavaScript, and Tailwind CSS, used daily by 10+ staff to run bookings, documents, and operations.
- Designed a reusable REST API layer of 25+ endpoints for CRUD operations, form handling, and document upload and management; implemented role-based access for users and administrators.
- Integrated the CRM with Microsoft Azure cloud services and handled deployment, debugging, and production fixes for a live business platform.
- Designed a normalized MySQL schema and ETL pipelines for staff, booking, and financial data; identified performance bottlenecks and database inefficiencies, cutting page load time by 40%.
- Built a KPI dashboard tracking revenue, margins, and booking volumes, and the company's public-facing website.

Software Web Development Intern

Main Flow Services and Technologies, Gurugram (Intern ID: 11787)

📅 May 2024 – Jul 2024

- Developed web applications using Python, JavaScript, SQL, HTML, and CSS across frontend and backend tasks, integrating components with SQL databases.
- Contributed features, bug fixes, and workflow improvements using Git and GitHub for version control and collaboration.

PROJECTS

Distributed Payment Wallet

Java | Spring Boot | PostgreSQL | Kafka | Redis

📅 2026

- Built a PayPal-style wallet system managing user wallets, internal ledger balances, and peer-to-peer transfers.
- Implemented the Saga pattern for distributed transactions across sharded PostgreSQL databases, with zero balance mismatches across 1,000+ concurrent transfers in load tests.
- Used PostgreSQL for strict ACID guarantees, Redis for locking concurrent balance updates, and Kafka for event-driven transaction auditing.

Enterprise Knowledge Intelligence Platform (RAG)

Python | FastAPI | PostgreSQL + pgvector | OpenAI API

📅 2026

- Productized an enterprise RAG platform for querying 500+ documents, SOPs, and knowledge articles using embeddings, vector search, and retrieval-augmented generation, exposed through FastAPI REST endpoints.
- Implemented retrieval with source citations and achieved 90%+ answer accuracy on a 100-query evaluation set.

Scalable Booking System with Dynamic Pricing

Python | FastAPI | Redis (Redlock) | PostgreSQL | Docker

📅 2026

- Built a high-concurrency seat booking platform, similar to airline and event ticketing systems, that handled 1,000+ concurrent requests in load tests without overselling a single seat.
- Solved double-booking using Redis distributed locks (Redlock) with optimistic and pessimistic database locking.
- Built a dynamic pricing engine that adjusts prices based on real-time demand and remaining inventory; containerized services with Docker.

AI Incident Resolution Agent

Python | OpenAI API | RAG | Tool Calling

📅 2026

- Built an AI agent that retrieves similar tickets, knowledge articles, and SOPs, then generates diagnoses and recommended resolutions using tool calling.
- Added human approval before automated actions; reached 90%+ resolution accuracy with 2–3s average latency.

Employee & Payroll Management System

Java | Spring Boot Microservices | Spring Security | MySQL

📅 2026

- Designed microservices for Auth, Employee, Payroll, Attendance, Leave, and Department with JWT, Spring Security, and RBAC for four user roles.
- Built REST APIs with Spring MVC and Spring Data JPA for payroll, payslip, leave, and attendance workflows on an indexed MySQL schema.

Transaction Fraud & Anomaly Detection

Python | SQL | Machine Learning

📅 2026

- Built a fraud detection pipeline combining SQL-based analysis and ML anomaly detection with class-imbalance handling, reaching 90%+ recall on fraudulent transactions, surfaced through an alert dashboard.

EDUCATION

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| • Master of Computer Applications (MCA) – AI & ML, Amity University Online | 2025 – Present |
| • Bachelor of Computer Applications (BCA) – AI & ML, Amity University, Noida | 2022 – 2025 |
| • Senior Secondary (CBSE) — 90%, Ryan International School, Noida | 2022 |

CERTIFICATIONS

- Data Science (Pursuing) — HCL GUVI · NPTEL Python Programming — IIT Madras
- AWS Cloud & Compute Fundamentals · Java Programming Certification
- React.js Frontend Development · NPTEL Innovation & Entrepreneurship — IIT Bombay

LANGUAGES

English — Professional Proficiency · Hindi — Native